Experiment : 1
Simulation and Comparative Performance Evaluation of Zigbee, 6LoWPAN and NB-IoT Communication Protocols for IoE Applications

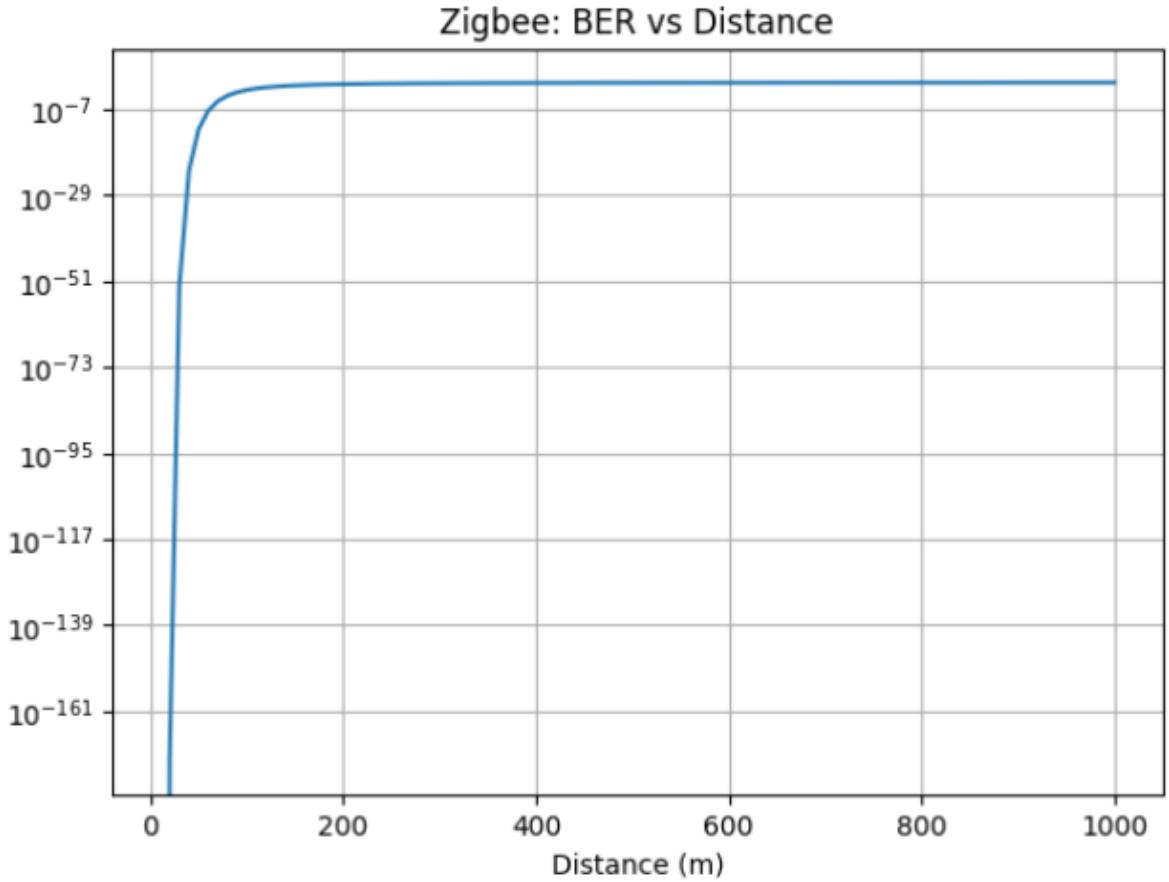
Objectives:To simulate and compare the performance of Zigbee, 6LoWPAN, and NB-IoT communication protocols for Internet of Everything (IoE) applications based on parameters such as data rate, latency, energy consumption, communication range, and reliability

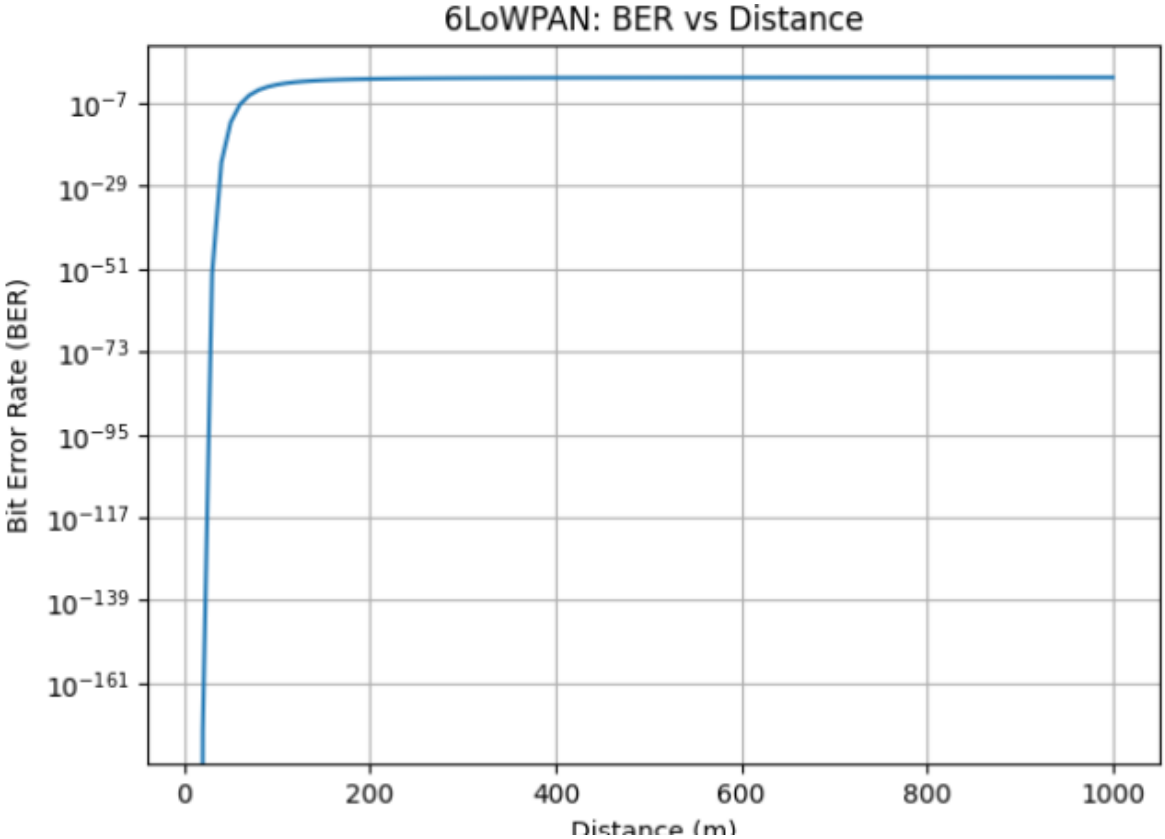
Procedure :

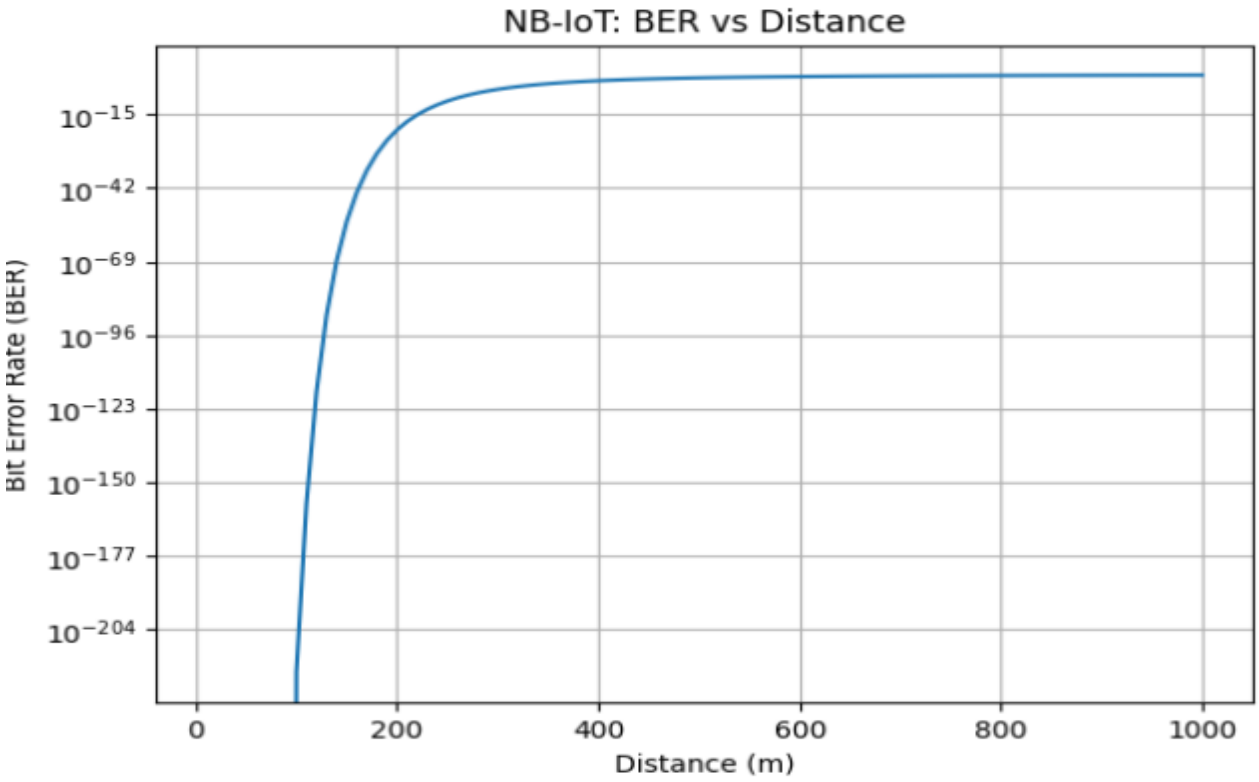
1. Select a suitable simulation tool and configure the network environment for Zigbee, 6LoWPAN, and NB-IoT protocols.
2. Create the network topology by configuring nodes, gateways, and communication links for each protocol.
3. Simulate data transmission between IoE devices using the three communication protocols under similar conditions.
4. Measure and record the performance parameters, including throughput, latency, energy consumption, and packet delivery ratio.
5. Compare the simulation results and identify the most suitable protocol for different IoE applications based on its performance.

Table column:

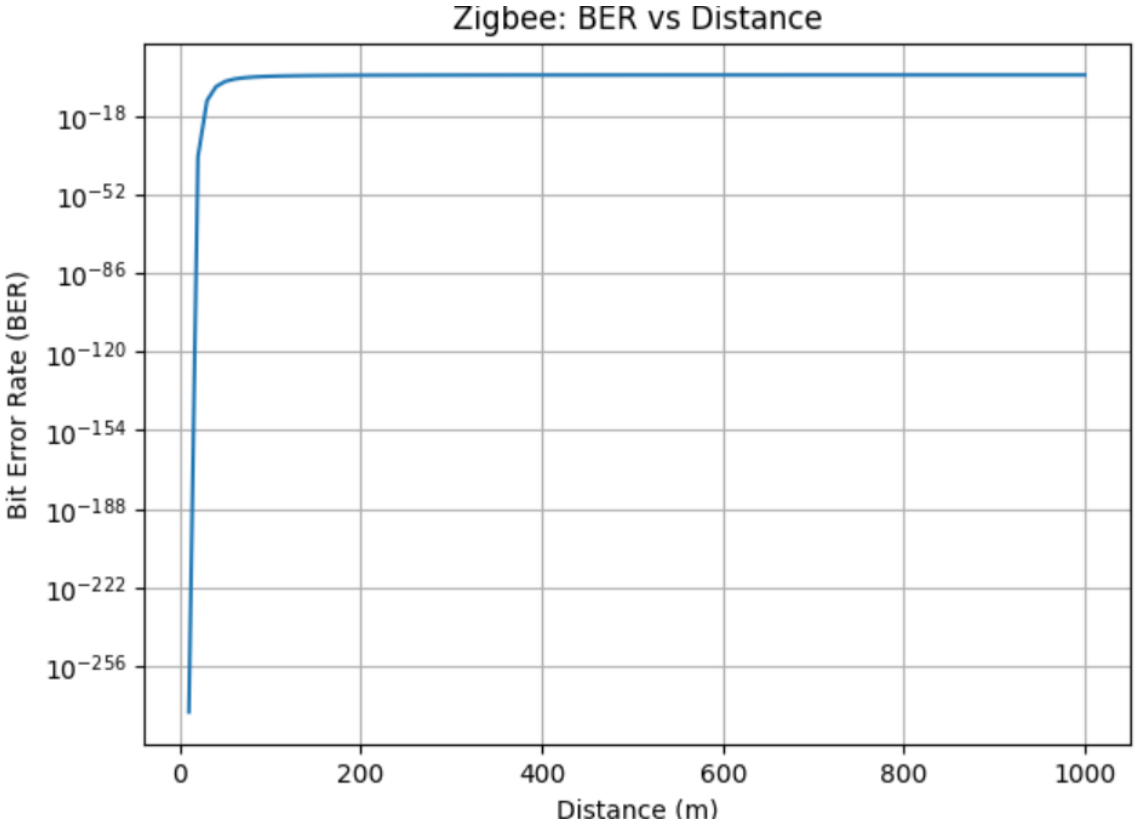
	ZIGBEE	6LoWPAN	NB-IoT
Pt(dBm)	0	0	23
PL0(dB)	40	40	31
N	3.0	3.0	3.5
N(dBm)	-105	-105	-105
BER	3.029e^-174	3.540e^-53	0.0036

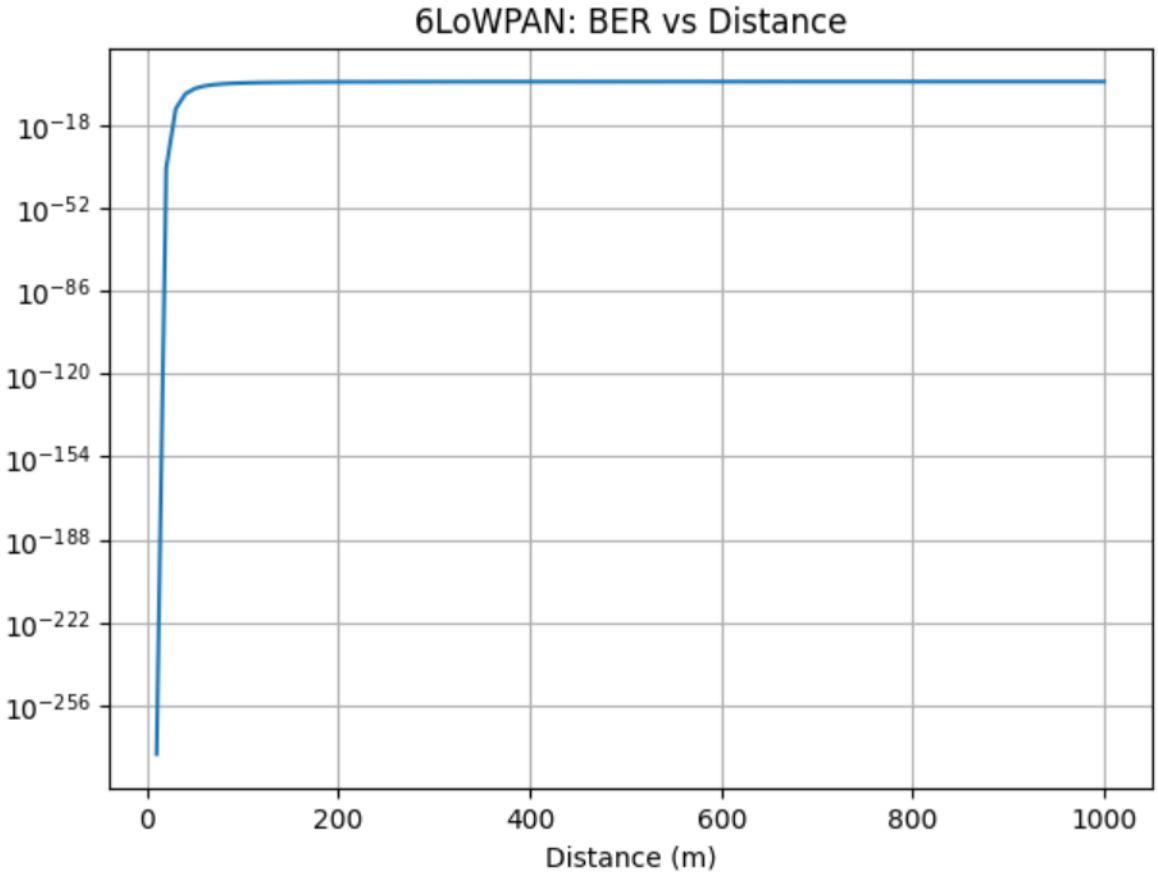

Zigbee: BER vs Distance
10^{-7}
10^{-29}
10^{-51}
10^{-73}
10^{-95}
10^{-117}
10^{-139}
10^{-161}
0
200
400
600
800
1000
Distance (m)

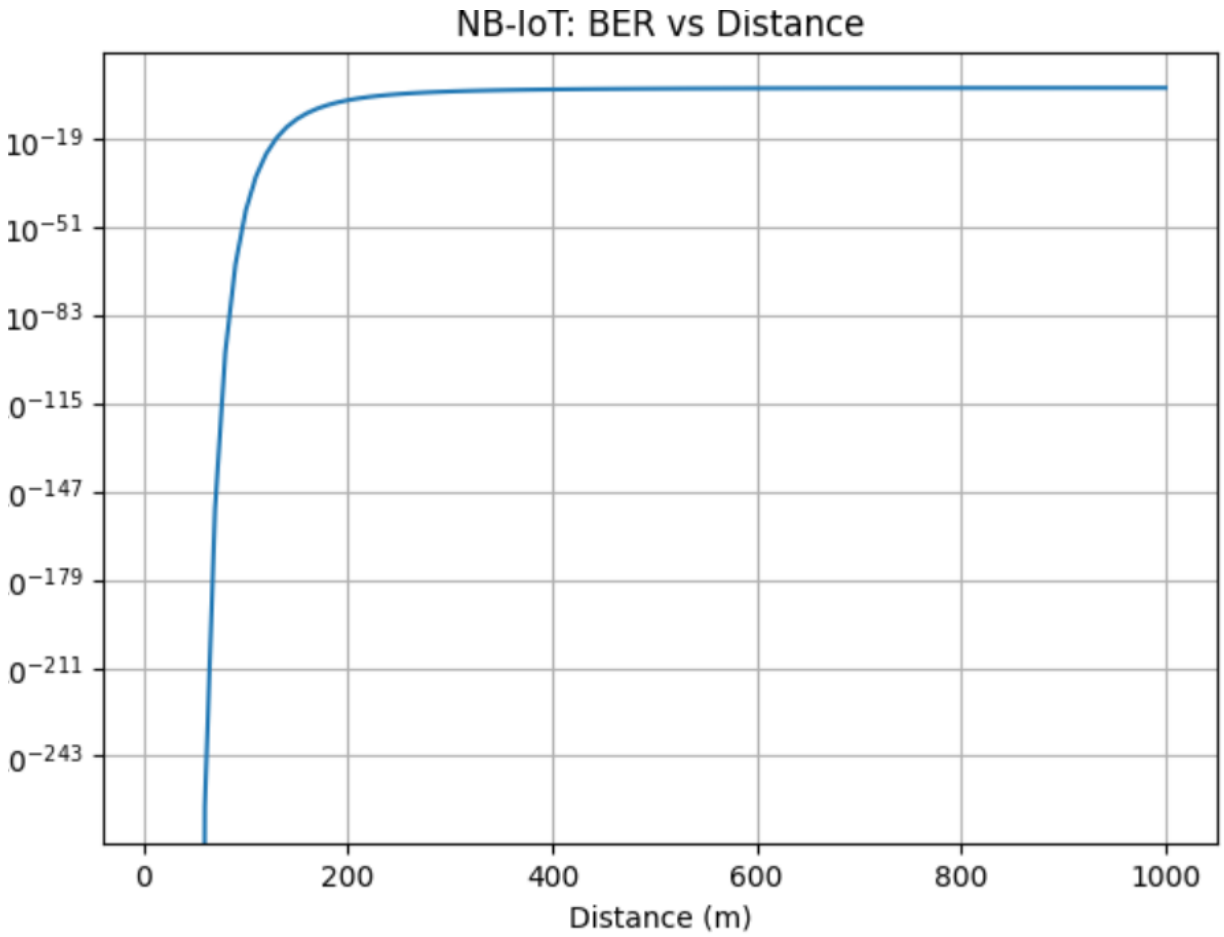

6LoWPAN: BER vs Distance
Bit Error Rate (BER)
10^{-7}
10^{-29}
10^{-51}
10^{-73}
10^{-95}
10^{-117}
10^{-139}
10^{-161}
0
200
400
600
800
1000
Distance (m)



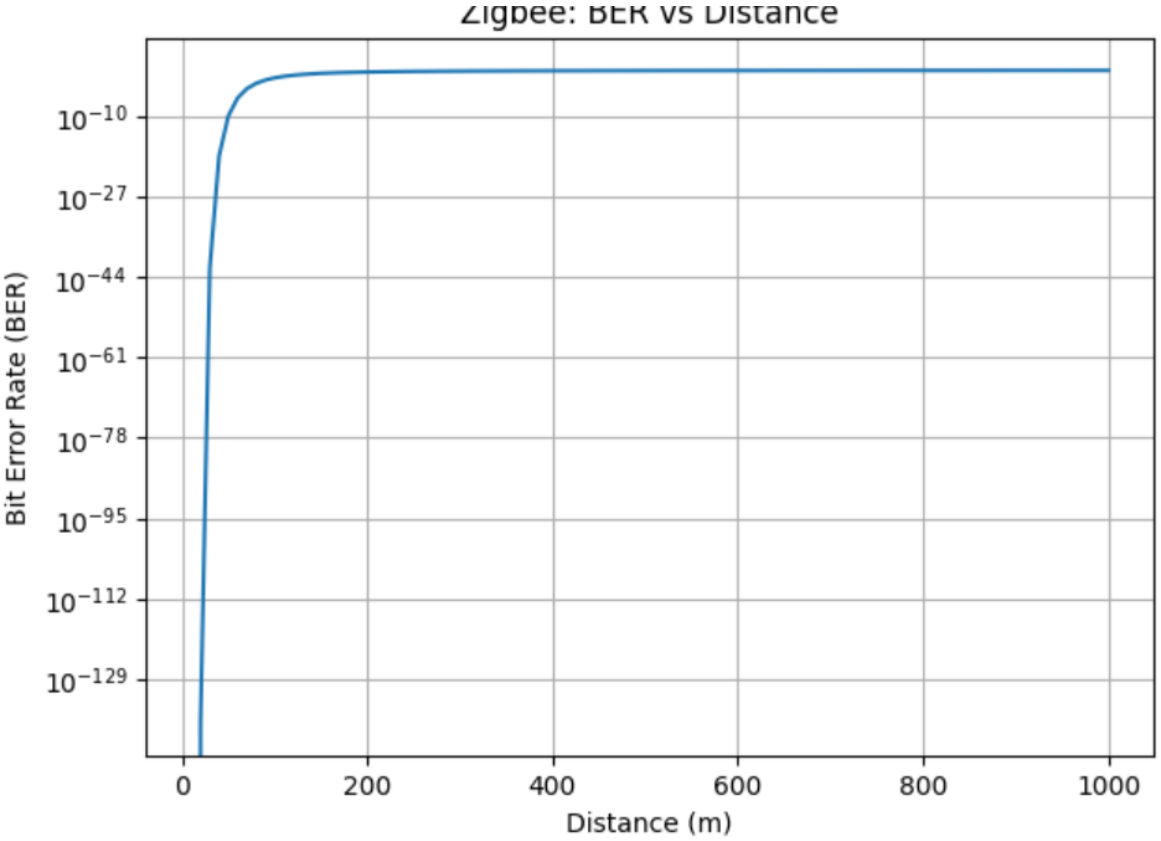
	ZIGBEE	6LoWPAN	NB-IoT
Pt(dBm)	0	0	23
PL0(dB)	40	40	31
N	3.0	3.0	3.5
N(dBm)	-98	-98	-98
BER	1.7642e^-36	4.0578^-12	0.115

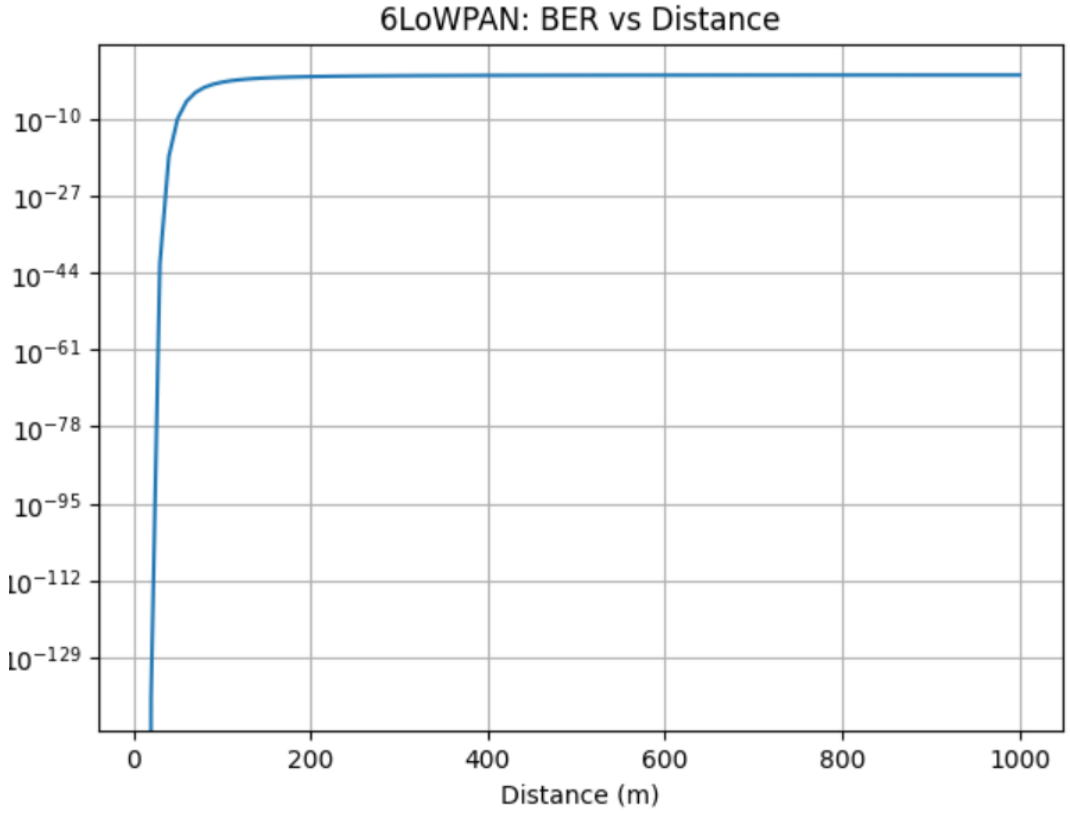

Zigbee: BER vs Distance
Bit Error Rate (BER)
10^{-18}
10^{-52}
10^{-86}
10^{-120}
10^{-154}
10^{-188}
10^{-222}
10^{-256}
0
200
400
600
800
1000
Distance (m)

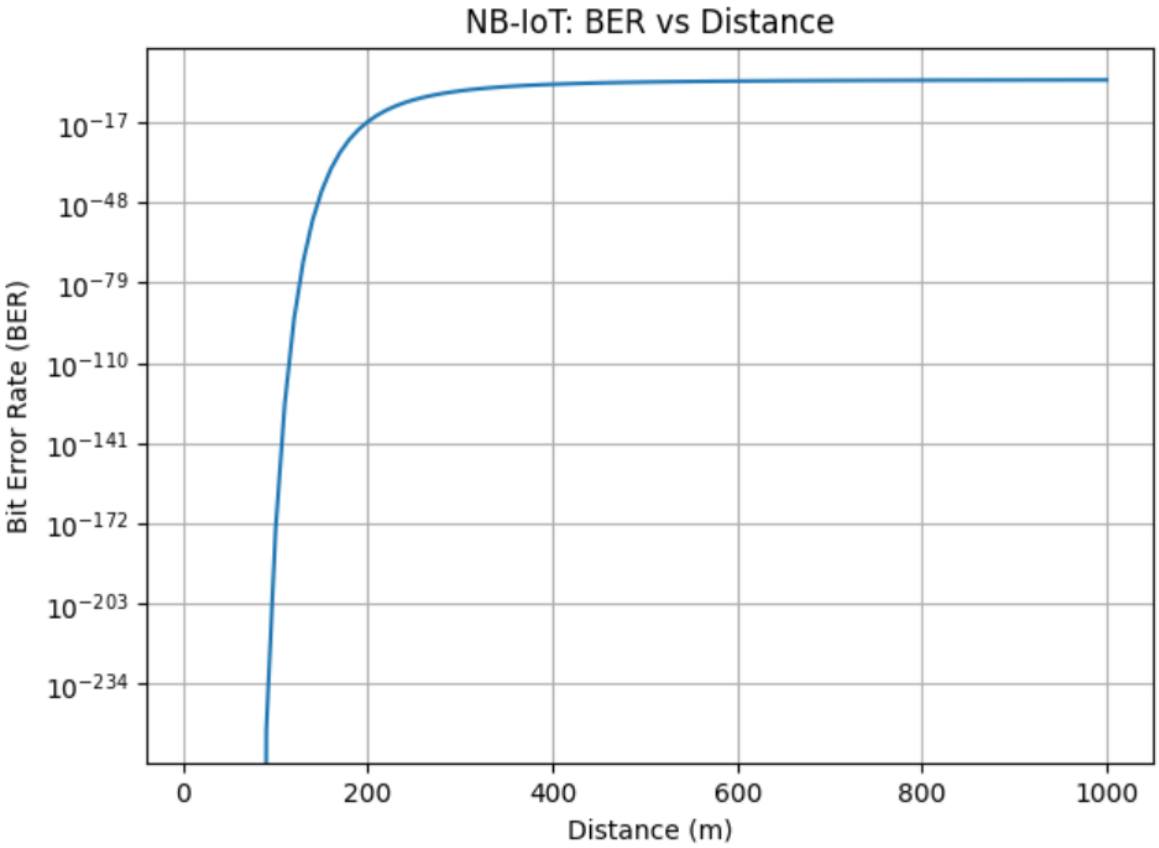

6LoWPAN: BER vs Distance
10^{-18}
10^{-52}
10^{-86}
10^{-120}
10^{-154}
10^{-188}
10^{-222}
10^{-256}
0
200
400
600
800
1000
Distance (m)



	ZIGBEE	6LoWPAN	NB-IoT
Pt(dBm)	0	0	23
PL0(dB)	40	40	31
N	3.0	3.0	3.5
N(dBm)	-104	-104	-104
BER	6.901^-136	1.148e^-42	0.0084


Zigbee: BER vs Distance
Bit Error Rate (BER)
10^{-10}
10^{-27}
10^{-44}
10^{-61}
10^{-78}
10^{-95}
10^{-112}
10^{-129}
0
200
400
600
800
1000
Distance (m)


6LoWPAN: BER vs Distance
10^{-10}
10^{-27}
10^{-44}
10^{-61}
10^{-78}
10^{-95}
10^{-112}
10^{-129}
0
200
400
600
800
1000
Distance (m)


NB-IoT: BER vs Distance
Bit Error Rate (BER)
10^{-17}
10^{-48}
10^{-79}
10^{-110}
10^{-141}
10^{-172}
10^{-203}
10^{-234}
0
200
400
600
800
1000
Distance (m)